2021有机化学与分析化学

时量180分钟　满分150分

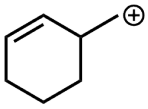
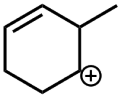
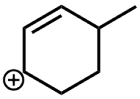
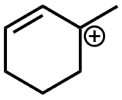
注意事项:

1. 答题前,考生先将自己的姓名、学号、考场号、座位号填写在答题卡上,将条形码准确粘贴在考生信息条形码粘贴区。
2. 考试结束后,将本试卷与答题纸一并收回。
3. 回答各个问题时,将答案写在答题纸上,写在本试卷上无效。

有机化学部分(共100分)

一、选择题（12 pts.）

1. 下列碳正离子中，最稳定的是

A. 
B. 
C. 
D. 

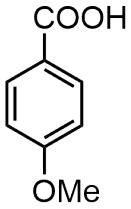
2. 与Lucas试剂反应，体系立刻变浑浊的是

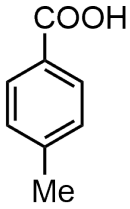
A.　正丁醇　　B.　酒精　　C.　苯甲醇　　D.　环己醇

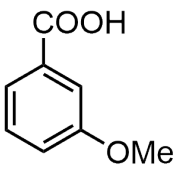
3. 下列试剂中，三苯甲基醚不稳定的是

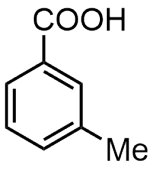
A.　碱　　B.　Grignard试剂　　C.　烷基锂　　D.　酸

4. 下列化合物中，酸性最强的是

A. 

B. 

C. 

D. 

5. 不能与HCN反应的化合物是

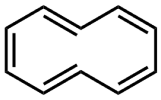
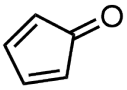
A.　环戊酮　　B.　苯乙酮　　C.　苯甲醛　　D.　2-丁酮

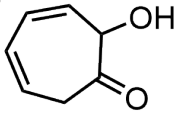
6. 下列化合物中，不能发生溴仿反应的是

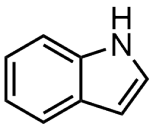
A. $BrCH_2COCH_2CH_3$　　B. $PhCOCH_3$

C. $CH_3CH(OH)CH_2CH_3$　　D. $PhCOC_2H_5$

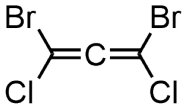
7. 下列化合物具有芳香性的是

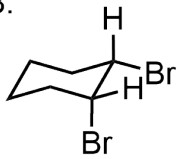
A. 
B. 

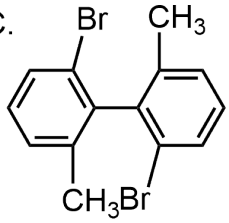
C. 

D. 

8. 下列化合物中，不具有手性的分子是

A. 

B. 

C. 

D.

9. 下列溶剂中，$C_2H_5O^-$的亲核性最强的是

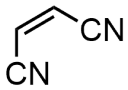
A. 二甲亚砜　　B. 水　　C. 三氟乙酸　　D. 乙醇

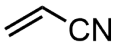
10. 下列化合物与对甲基苯磺酰氯反应后既不溶于酸又不溶于碱的是

A. 乙胺　　B. 四氢吡咯　　C. 环戊胺　　D. 三乙胺

11. 下列亲双烯体中，[4+2]环加成反应活性最高的是

A.

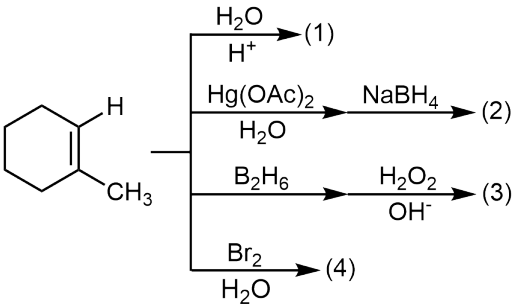
B. 

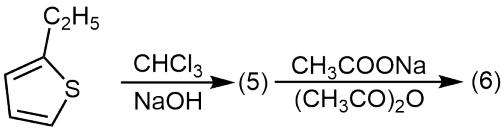
C.
D. 

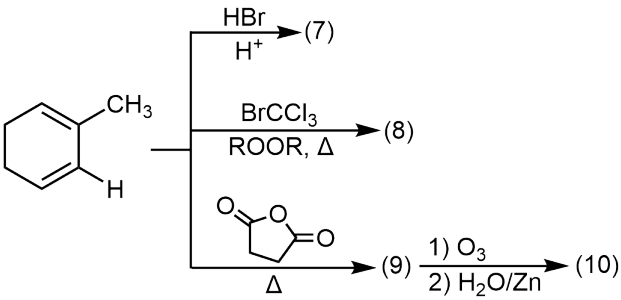
12. 吡啶发生磺化反应时，磺酸基主要进入

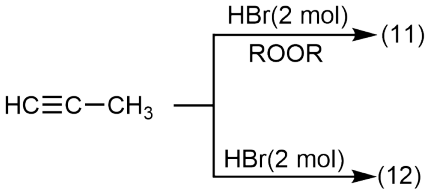
A. α位　　B. β位　　C. γ位　　D. α位和γ位

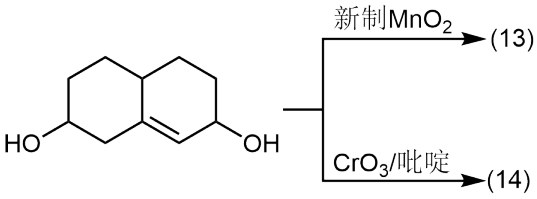
二、完成反应(32 pts.)

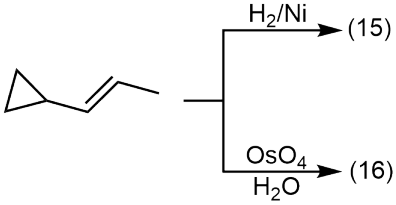
1. 

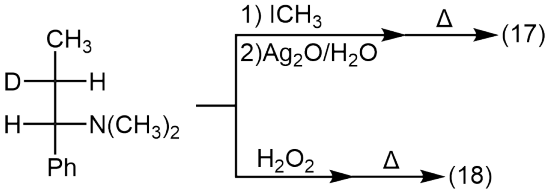
2. 
C2H5
S
CHCl3
NaOH
(5)
CH3COONa
(CH3CO)2O
(6)

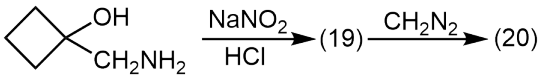
3. 
CH3
H
HBr
H+
(7)
BrCCl3
ROOR, Δ
(8)
O
O
O
Δ
(9)
1) O3
2) H2O/Zn
(10)

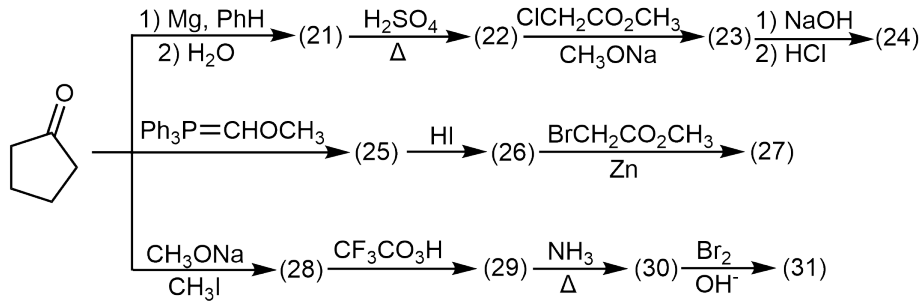
4. 
HC≡C−CH3
HBr(2 mol)
ROOR
(11)
HBr(2 mol)
(12)

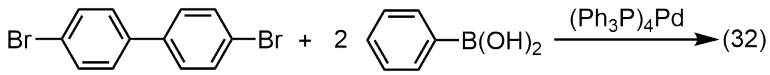
5. 
HO
OH
新制MnO2
(13)
CrO3/吡啶
(14)

6. 
H2/Ni
(15)
OsO4
H2O
(16)

7. 
CH3
D
H
H
N(CH3)2
Ph
1) ICH3
2)Ag2O/H2O
Δ
(17)
H2O2
Δ
(18)

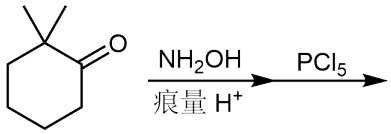
8. 
OH
CH2NH2
NaNO2
HCl
(19)
CH2N2
(20)

9. 

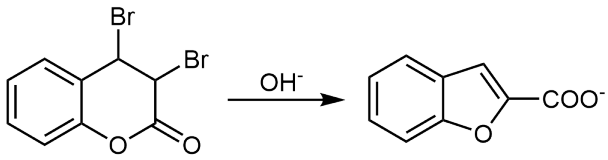
10. 

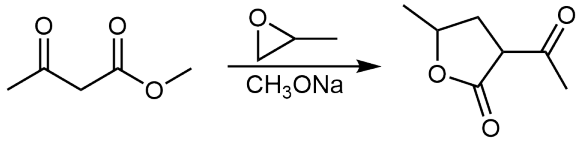
三、简答题(7 pts.)

1. 完成下列反应，给出其机理，并解释醛酮与羟胺反应生成肟需要用酸催化，却不能用强酸催化的原因。



四、机理题(14 pts.)

1. 

2. 

五、合成题(35 pts.)

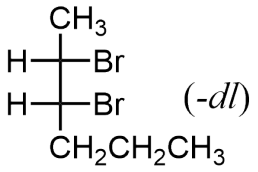
1. 以丙炔为唯一有机原料合成**A**

2. 以苯、小于等于4个碳的有机物合成**B**

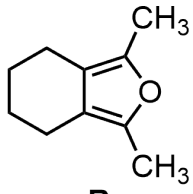
3. 以苯、小于等于2个碳的有机物合成**C**

4. 以乙炔、小于等于3个碳的有机物合成**D**

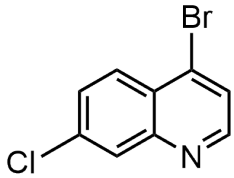
5. 以环己酮、小于等于3个碳的有机物合成**E**



A



B



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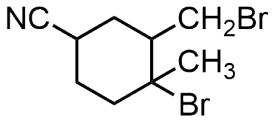

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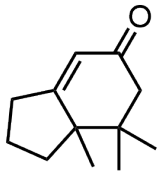

E

分析化学部分(共50分)

六、简答题(20 pts.)

1. 用最小刻度为0.01 g的台秤称取约5 g的物品，最多可记录多少位有效数字？如用来测定土壤水分，要求称量的相对误差不大于0.2%，至少应称取土壤试样多少克？

2. 写出下列溶液的质子条件式：
   (1) NH_4HCO_3水溶液质子条件式。
   (2) NaOH和NaAc混合水溶液的质子条件式（假定NaOH在水溶液中全部离解，其浓度为c_b）。
   (3) H_2NCH_2COOH水溶液的质子条件式。
   (4) HAc−NaAc缓冲溶液的质子条件式（假定HAc在缓冲溶液中浓度为c_a，NaAc在缓冲溶液中浓度为c_b）。

3. 若配制EDTA溶液的水中含有少量Pb^{2+}, Ca^{2+}，今在pH 5 ∼ 6用纯锌为基准物标定其浓度。(1)测定Al^{3+}（pH5 ∼ 6，以Zn^{2+}标准溶液返滴定过量的EDTA），其结果会怎样；(2)测定Ca^{2+}其结果会怎样。（指偏高、偏低或无影响）

4. 已知在1 mol/L HCl中，$\varphi^{\ominus}_{Fe^{3+}/Fe^{2+}} = 0.68\,V$；$\varphi^{\ominus}_{Sn^{4+}/Sn^{2+}} = 0.14\,V$，问以$Fe^{3+}$滴定$Sn^{2+}$至99.9%、100%、100.1%时的电位分别为多少？

5. 请简要回答为什么莫尔(Mohr)法适用的酸度范围是$pH = 6.5 - 10.5$？

七、计算题(30 pts.)

1. 以0.1000 mol/L NaOH溶液滴定0.2000 mol/L HA和0.2000 mol/L HB的混合溶液。求：(1)判断是否能选择性滴定HA，并计算滴定至化学计量点时体系的pH值；(2)若终点时pH值比化学计量点时高0.5个pH单位，终点误差是多少？

   [已知：HA的$K_a = 10^{-3.74}$，HB的$K_a = 10^{-9.26}$]

2. 在pH = 5.0时，以二甲酚橙作指示剂用2.000×10^{-2} mol/L EDTA滴定浓度皆为2.000×10^{-2} mol/L的Pb^{2+}、Zn^{2+}、Ca^{2+}混合液中的Pb^{2+}，用邻二氮菲(phen)掩蔽Zn^{2+}，已知终点时phen游离浓度为$10^{-2.0}$ mol/L，(1)问Zn^{2+}被掩蔽的效果；(2)计算终点误差（忽略加掩蔽剂后的体积变化）。

   [已知：$\lg K_{\mathrm{PbY}} = 18.0$，$\lg K_{\mathrm{ZnY}} = 16.5$，$\lg K_{\mathrm{CaY}} = 10.7$，Zn−phen络合物的$\lg \beta_3 = 17.6$，pH = 5.0时$\mathrm{pPb_{ep}}$(二甲酚橙)= 7.0、$\lg \alpha_{Y(\mathrm{H})} = 6.6$]

3. 计算Ag_2S在pH = 9.0的NH_3-NH_4Cl缓冲溶液（$[NH_3] = 0.10$ mol/L），S^{2-}总浓度为0.10 mol/L溶液中的溶解度。

   [H_2S：$K_{a1} = 1.3\times10^{-7}$，$K_{a2} = 7.1\times10^{-15}$；$Ag^+-NH_3$络合物$\lg \beta_1, \lg \beta_2$依次为$3.4, 7.4$；$K_{\mathrm{sp}}(Ag_2S) = 2.0\times10^{-50}$]

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注意事项:

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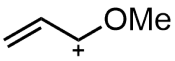
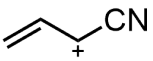
2. 考试结束后,将本试卷与答题纸一并收回。
3. 回答各个问题时,将答案写在答题纸上,写在本试卷上无效。

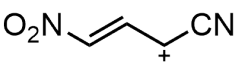
有机化学部分(共100分)

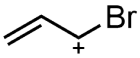
一、选择题（15 pts.）

1. 下列碳正离子中，最稳定的是

A. 

B. 

C. 

D. 

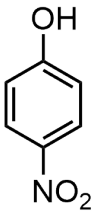
2. 不能与饱和亚硫酸氢钠溶液反应的化合物是

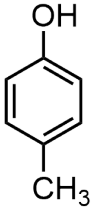
A.　环戊酮　　B.　葡萄糖　　C.　苯甲醛　　D.　2-丁酮

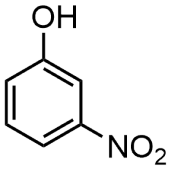
3. 不能发生碘仿反应的化合物是

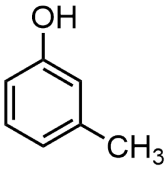
A.　苯乙酮　　B.　乙醇　　C.　乙醚　　D.　乙酰乙酸乙酯

4. 下列化合物中，酸性最强的是

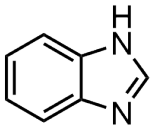
A. 

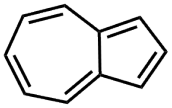
B. 

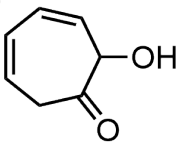
C. 

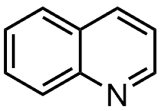
D. 

5. 下列化合物中，不具有芳香性的是

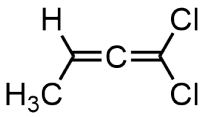
A. 

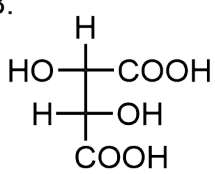
B. 

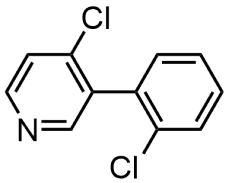
C. 

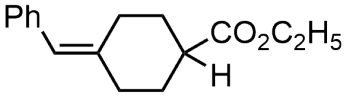
D. 

6. 下列化合物中，具有手性的分子是

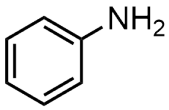
A. 

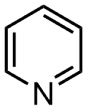
B. 

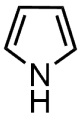
C. 

D. 

7. 下列化合物中，碱性最强的是

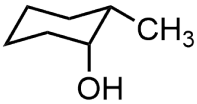
A. 

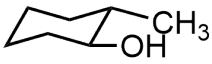
B. 

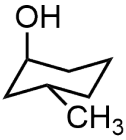
C. 

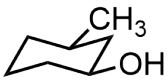
D.

8. 下列化合物构象最稳定的是

A. 

B. 

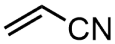
C. 

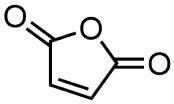
D. 

9. 下列亲双烯体中，[4+2]环加成反应活性最高的是

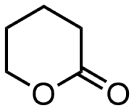
A.

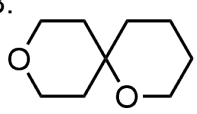
B.

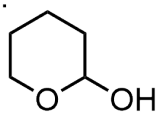
C. 

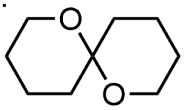
D. 

10. 下列化合物中，不能被稀酸水解的是

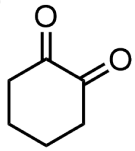
A. 

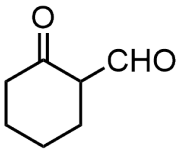
B. 

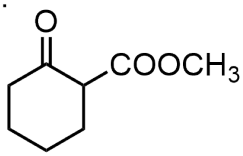
C. 

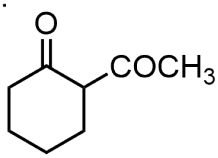
D. 

11. 在四氯化碳中，下列化合物烯醇式含量最高的是

A. 

B. 

C. 

D. 

12. 下列试剂不能把羰基还原为亚甲基的是

A. Zn(Hg),HCl　　B. $KOH/O(CH_2CH_2OH)_2/NH_2NH_2$

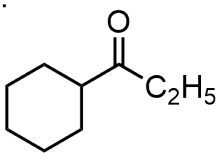
C. $LiAlH_4$　　D. $HSCH_2CH_2SH/Pt$

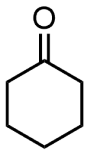
13. 下列试剂中，亲核性最好的是

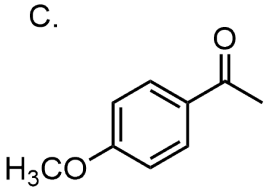
A. $CH_3CH_2COO^-$　　B. $CH_3CH_2S^-$

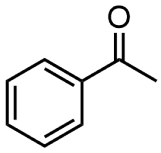
C. $CH_3CH_2O^-$　　D. CH_3CH_2SH

14. 下列羰基化合物与CH_3MgI反应时，慢反应一步活化能最低的是

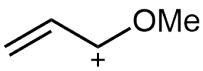
A. 

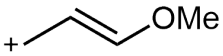
B. 

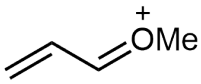
C. 

D. 

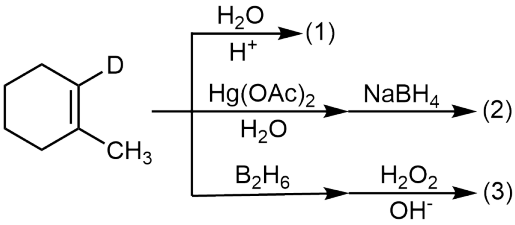
15. 下列共振式中对杂化体贡献最大的是

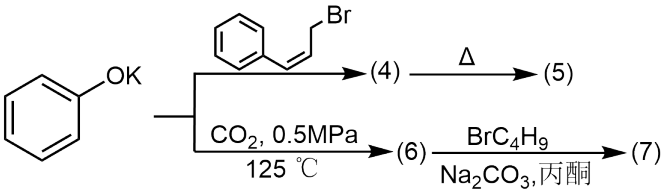
A. 

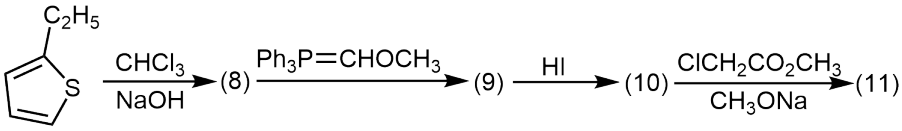
B. 

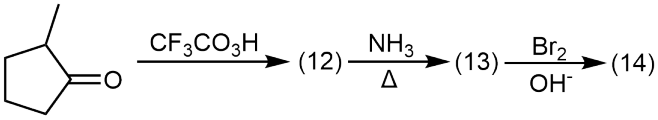
C. 

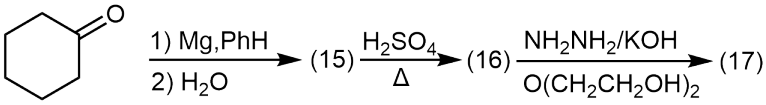
二、完成反应(30 pts.)

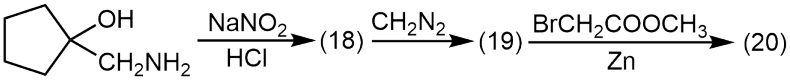
1. 

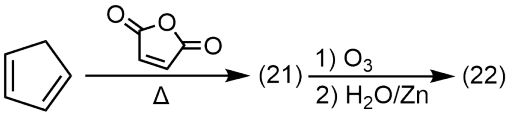
2. 

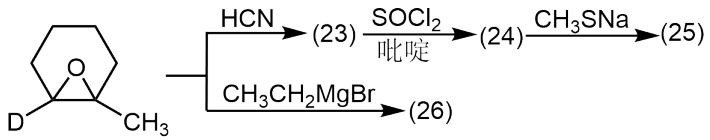
3. 

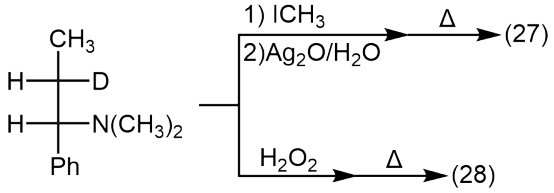
4. 

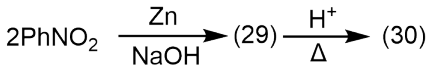
5. 

6. 

7. 

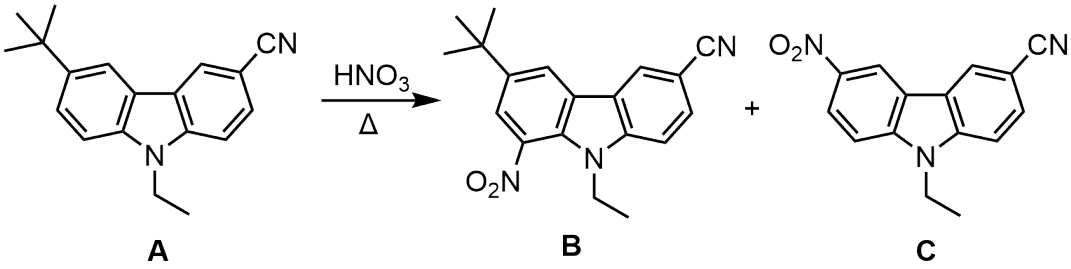
8. 

9. 

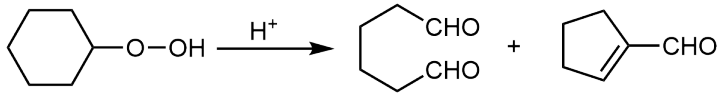
10. 

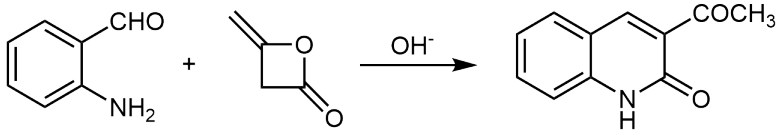
三、简答题(6 pts.)

1. 有人想用化合物A通过硝酸硝化得到化合物B，但是却得到了化合物C为主的产物。试解释这一现象。



四、机理题(14 pts.)

1. 

2. 

五、合成题(35 pts.)

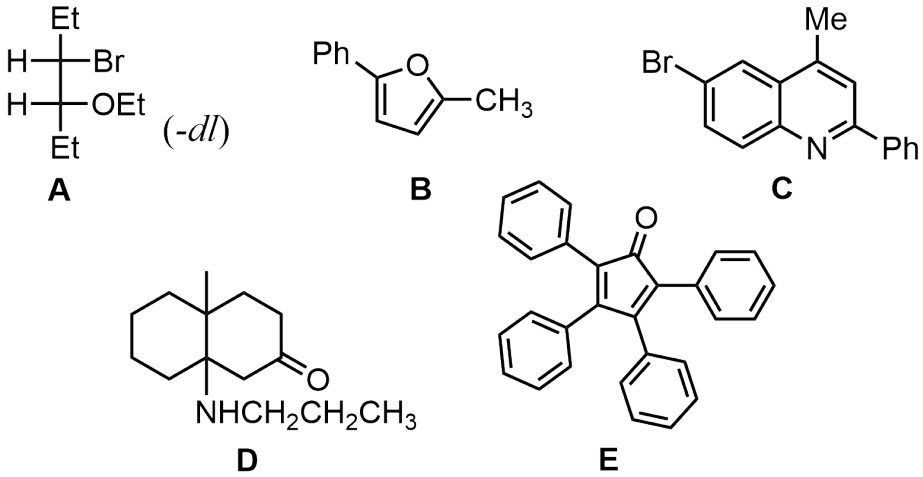
1. 以乙炔为唯一有机原料合成**A**

2. 以苯、小于等于2个碳的有机物合成**B**

3. 以苯、小于等于3个碳的有机物合成**C**

4. 以环己酮、小于等于3个碳的有机物合成**D**

5. 以苯、小于等于3个碳的有机物合成**E**



分析化学部分(共50分)

六、简答题（每小题4分）

1. 络合滴定中为什么要加入缓冲溶液?请说明原因。

2. 写出下列水溶液的质子平衡条件式

   (1) $(NH_4)_2CO_3$

   (2) 0.2 mol/L H_2SO_4

3. 简述EDTA的$\lg \alpha_{Y(H)}$随pH的变化趋势，如只考虑酸效应，则金属离子M与EDTA络合物的条件稳定常数K'_{MY}的表达式是什么?

4. 已知半反应$VO_2{}^+ + 2\,H^+ + e^- = VO^{2+} + H_2O$的标准电极电位$\varphi^{\ominus} = 1.00$ V,计算pH = 2.00时，溶液中电对$VO_2{}^+/VO^{2+}$的条件电位$\varphi^{\ominus\prime}$。

5. 用甲醛法测得某铵盐中氮的质量分数(%)为5.15, 5.32, 5.22, 5.25,计算含氮量在置信度95%时总体平均值的置信区间。[已知：$t_{0.05,3} = 3.18$]

七、计算题（每小题10分，共30分）

1. 用0.1000 mol/L NaOH溶液滴定20.00 mL 0.05000 mol/L H_2A,计算：

(1) 化学计量点前0.1%的pH;
(2) 化学计量点的pH;
(3) 化学计量点后0.1%的pH

[已知：H_2A的$pK_{a1} = 1.22$,$pK_{a2} = 4.19$]

2. 以0.02000 mol/L EDTA滴定浓度均为0.02000 mol/L Zn^{2+}、Ca^{2+}混合液中的Zn^{2+},溶液pH为5.0,若以二甲酚橙为指示剂，终点误差多大?

   [已知：pH = 5.0时$\lg \alpha_{Y(H)} = 6.6$,pZn_{ep}(二甲酚橙)= 4.8;$\lg K_{ZnY} = 16.5$,$\lg K_{CaY} = 10.7$]

3. 计算AgI在0.01 mol/L $Na_2S_2O_3$和0.01 mol/L KI混合溶液中的溶解度。

   [已知：Ag^+与I^-络合$\lg \beta_1 \sim \lg \beta_3$依次为6.58, 11.74, 13.68;$Ag^+$与${S_2O_3}^{2-}$络合$\lg \beta_1 \sim \lg \beta_3$依次为8.82, 13.46, 14.15; $K_{sp}(AgI) = 9.3 \times 10^{-17}$]

2023有机化学与分析化学

时量180分钟　满分150分

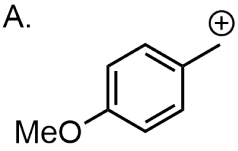
注意事项:

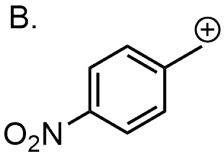
1. 答题前,考生先将自己的姓名、学号、考场号、座位号填写在答题卡上,将条形码准确粘贴在考生信息条形码粘贴区。
2. 考试结束后,将本试卷与答题纸一并收回。
3. 回答各个问题时,将答案写在答题纸上,写在本试卷上无效。

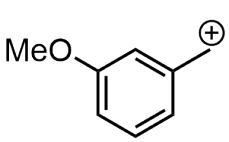
有机化学部分(共100分)

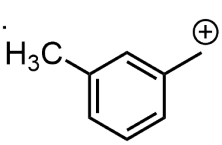
一、选择题（15 pts.）

1. 下列碳正离子中，最稳定的是

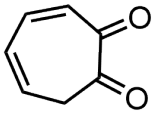
A. 

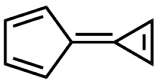
B. 

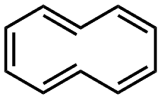
C. 

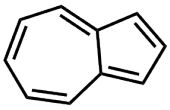
D. 

2. 下列化合物不具有芳香性的是

A. 

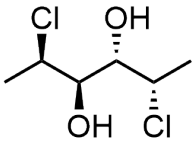
B. 

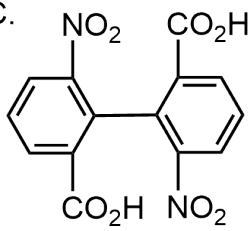
C. 

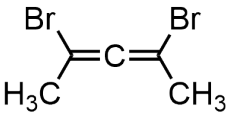
D. 

3. 下列化合物中，不具有手性的分子是

A.

B. 

C. 

D. 

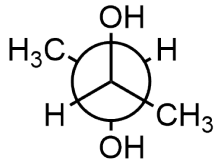
4. 下列试剂中，三苯甲基醚不稳定的是

A. 碱　　B. Grignard试剂　　C. 烷基锂　　D. 酸

5. 下列试剂中，酸性最强的是

A. 三氯乙酸　　B. 乙酸　　C. 氯乙酸　　D. 溴乙酸

6. 下列化合物有其它三项结构不同的是

A.


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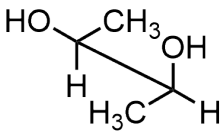

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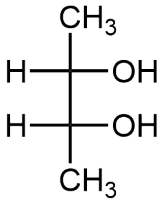

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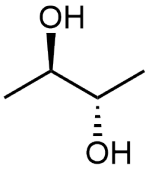
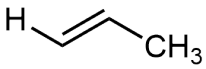


7. 下列刻意绘出的端氢中，酸性最弱的是

A.


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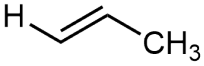

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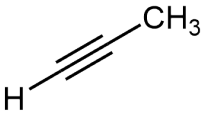
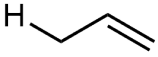
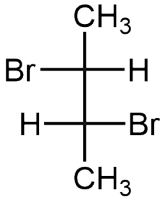


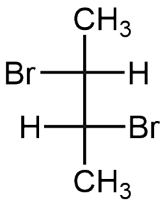
D.


8. 下图所示分子发生单氯代反应所得同分异构体数目为



A. 3　　B. 4　　C. 5　　D. 6

9. 在酸性条件下，2-甲基环己酮与等摩尔的溴单质发生反应时，将发生在哪个位置



A. 1　　B. 2　　C. 3　　D. 4

10. 等摩尔甲醛和苯甲醛经浓碱处理将生成

A. 甲醇+苯甲醇　　B. 甲酸负离子+苯甲酸负离子

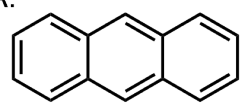
C. 苯甲酸负离子+甲醇　　D. 甲酸负离子+苯甲醇

11. 下列化合物的水解反应速率最快的是

A. $CF_3COOC_2H_5$　　B. $CH_3COOC_2H_5$

C. $CH_3CH_2COOC_2H_5$　　D. $ClCH_2COOC_2H_5$

12. 下列化合物中，不能和乙烯发生[4+2]环加成反应的是

A.


B.

C.

D.

13. 下列化合物不能与HCN发生反应的是

A. $CH_3CH(OH)CH_2CH_3$　　B. $CH_2ICOCH_2CH_3$

C. CH_3COCH_2Ph　　D. CH_3OCOCH_2CHO

14. 下列化合物不能发生碘仿反应的是

A. 环己酮　　B. 苯乙酮　　C. 呋喃甲醛　　D. 2-丁酮

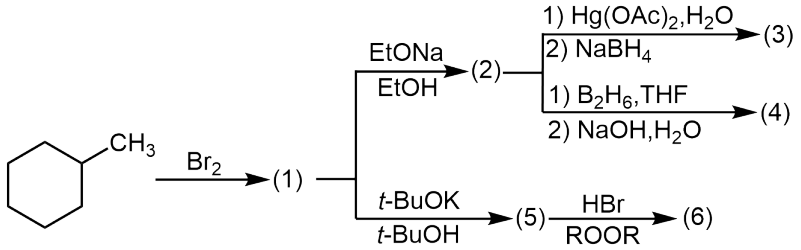
15. 下列化合物中，烯醇式含量最高的是

A. $CH_3COCH_2CONH_2$　　B. $CH_3COCH_2COCH_3$

C. CH_3COCH_2CHO　　D. $CH_3COCH_2COOCH_3$

二、完成反应(30 pts.)

1.



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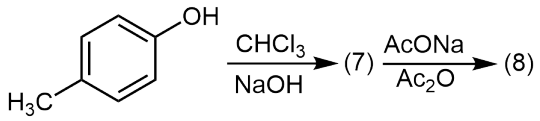

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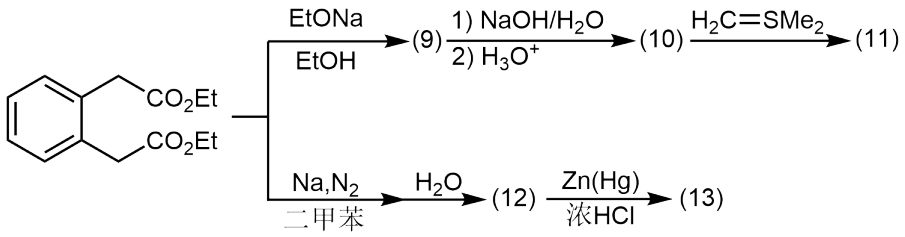

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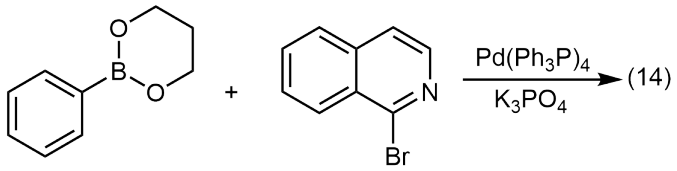

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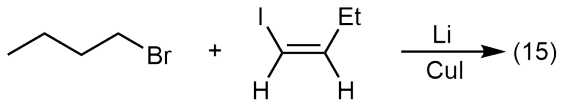
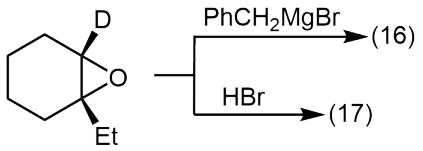
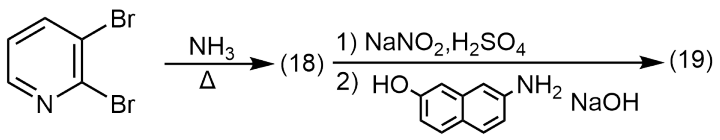
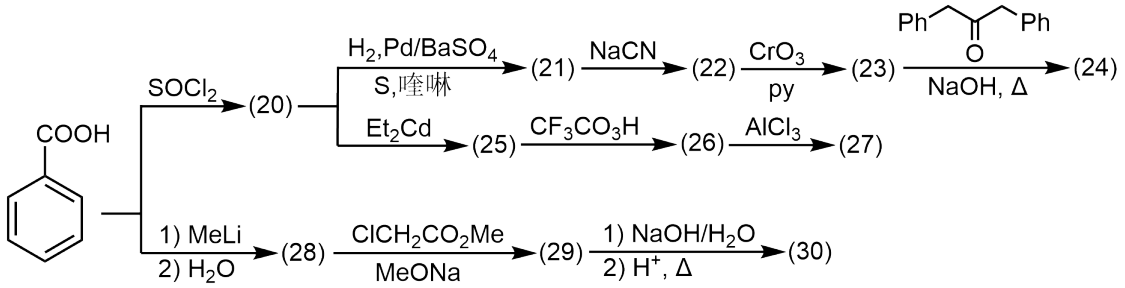


6. 

7. 

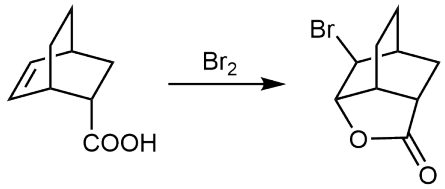
8. 

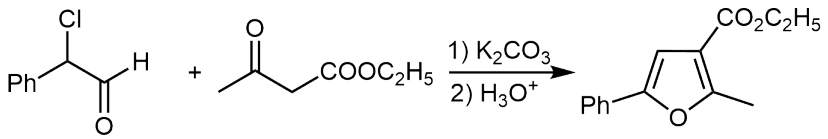
三、简答题(6 pts.)

1. 试解释下列化合物发生硝化反应的相对速率

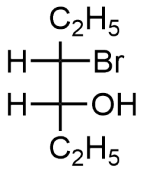
化合物	相对反应速率
硝基苯	6×10^{-8}
氯苯	0.033
苯	1
苯酚	1000

四、机理题(14 pts.)

1. 

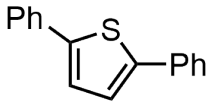
2. 

五、合成题(35 pts.)

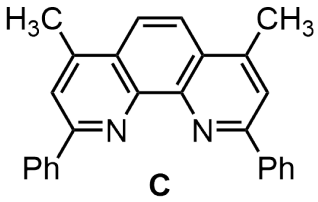


(-*dl*)

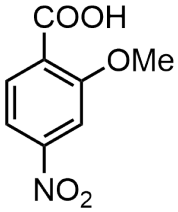
A



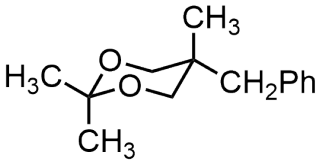
B



C



D



E

1. 以乙炔为唯一的有机原料合成**A**

2. 以苯和小于等于2个碳的有机化合物为原料合成**B**

3. 以苯或甲苯和小于等于3个碳的有机化合物为原料合成**C**

4. 以苯或甲苯和小于等于2个碳的有机化合物为原料合成**D**

5. 以苯和小于等于3个碳的有机化合物为原料合成**E**

分析化学部分(共50分)

六、简答题

1. 根据有效数字运算规则，计算下列算式结果

   (1) $11.05 + 1.3153 + 1.225 + 25.0678$
   (2) 将 3.451 保留两位有效数字
   (3) 将 15.3748 保留四位有效数字

2. 已知半反应 $VO_2{}^+ + 2\,H^+ + e^- = VO^{2+} + H_2O$ 的标准电极电位 $\varphi^{\ominus} = 1.00\,V$，计算 $pH = 2.00$ 时，溶液中电对 $VO_2{}^+/VO^{2+}$ 的条件电位 $\varphi^{\ominus\prime}$。

3. 莫尔法所用的指示剂是什么？请说明莫尔法通常在中性或弱碱性溶液中滴定的原因。

4. 用 EDTA 在弱碱性溶液中滴定 Cu^{2+} 时，常在 NH_3-$NH_4{}^+$溶液中进行，其作用是什么？

5. 求 CaF_2 在 0.01 mol/L HCl 溶液中的溶解度（忽略沉淀溶解消耗的酸），已知 $K_a = 10^{-3.18}$，$K_{\mathrm{sp}} = 10^{-10.57}$。

七、计算题

1. 用 0.10 mol/L 的 NaOH 溶液滴定 0.10 mol/L 的 HCl 溶液和 0.20 mol/L 的一元弱酸 HA 的混合溶液（已知 $K_a = 5.8 \times 10^{-10}$）
   (1) 计算化学计量点时溶液的 pH
   (2) 若滴定终点 pH 比化学计量点的 pH 高 0.5 个 pH 单位，计算终点误差

2. 浓度均为 0.0100 mol/L 的 Zn^{2+}, Cd^{2+} 混合溶液，加入过量 KI，使终点时游离 I^- 浓度为 1.0 mol/L，在 pH = 5.0 时，以二甲酚橙做指示剂，用等浓度的 EDTA 滴定其中的 Zn^{2+}，计算终点误差。
   已知：$\lg K_{\mathrm{ZnY}} = 16.50$，$\lg K_{\mathrm{CdY}} = 10.7$，$Cd^{2+}$ 与 I^- 形成的络合物的 $\lg \beta_1 - \lg \beta_4$ 依次为 $2.10, 3.43, 4.49, 5.41$；pH = 5.0 时，$\mathrm{pZn_{ep}}$(二甲酚橙) = 4.8，$\lg \alpha_{Y(\mathrm{H})} = 6.453$。

3. 用一种新方法测定试样中的锌（%），平行测定了 6 次，测得结果如下（%）：$1.56, 1.58, 1.62, 1.60, 1.68, 1.65$，试求：
   (1) 平均值的置信区间（$P = 95\%$）
   (2) 若已知试样中锌的标准值为 1.68%，试问采用新方法后是否引起系统误差（显著性水平 0.05）

   已知：

f	4	5	6
$t_{0.05}$	2.78	2.57	2.45

2024有机化学与分析化学

时量180分钟　满分150分

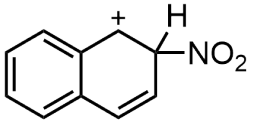
注意事项:

1. 答题前,考生先将自己的姓名、学号、考场号、座位号填写在答题卡上,将条形码准确粘贴在考生信息条形码粘贴区。
2. 考试结束后,将本试卷与答题纸一并收回。
3. 回答各个问题时,将答案写在答题纸上,写在本试卷上无效。

有机化学部分(共100分)

一、选择题（15 pts.）

1. 下列共振式中贡献最大的是

A.


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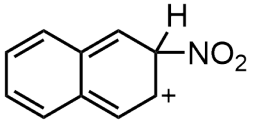

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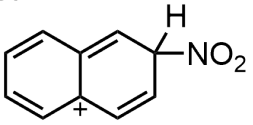

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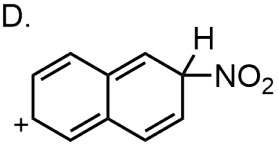
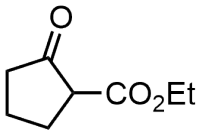


2. 己二酸二乙酯在二甲苯、钠存在下生成

A.


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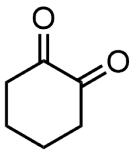

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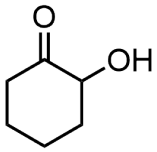

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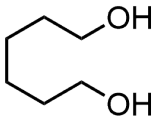

3. 下列化合物亲核加成活性最高的是

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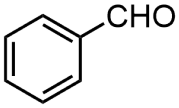

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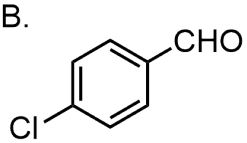

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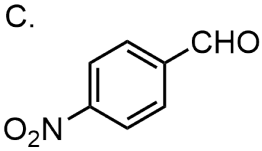

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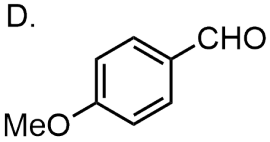


4. 浓硫酸能除去苯中少量噻吩的原因是

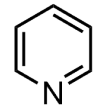
A. 苯与浓硫酸互溶

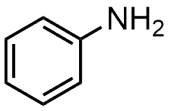
B. 噻吩与浓硫酸互溶

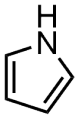
C. 噻吩与浓硫酸发生亲电反应活性更高

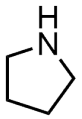
D. 苯与浓硫酸发生亲电反应活性更高

5. 将下列化合物的碱性进行排序

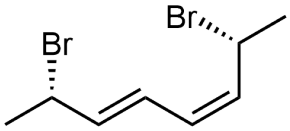
A. 

B. 

C. 

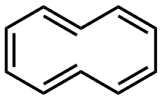
D. 

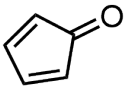
6. 判断下列化合物的构型

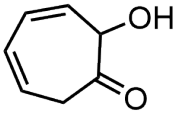


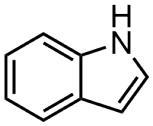
A. 2S,3E,5Z,7R　　B. 2R,3E,5S,7Z　　C. 2S,3Z,5R,7E　　D. 2R,3Z,5E,7S

7. 下列化合物不是内消旋体的是

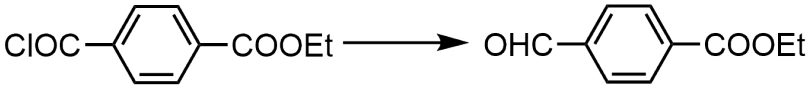
A. 

B. 

C. 

D. 

8. 给出如下反应的可能反应条件



A. $LiAlH_4$　　B. $H_2/Pd-C$

C. $H_2/Pd/BaSO_4$,S,Quinoline　　D. Na/EtOH

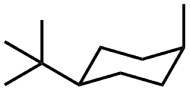
9. 不能与HCN反应的化合物是

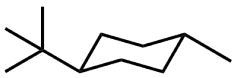
A. 环戊酮　　B. 苯乙酮　　C. 苯甲醛　　D. 2-丁酮

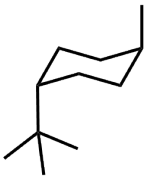
10. 下列化合物碱性最强的是

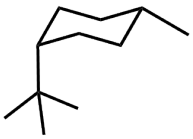
A. CH_3ONa　　B. CH_3SH　　C. $(CH_3)_3CONa$　　D. $(CH_3)_2CHONa$

11. 顺-1-甲基-4-叔丁基最稳定的构象是

A. 

B. 

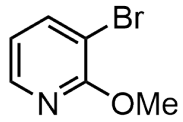
C. 

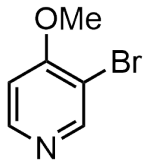
D. 

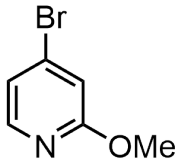
12. (S)-苯基仲丁基酮在碱中长时间放置会

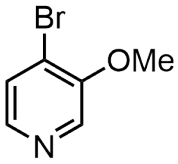
A. 构型保持　　B. 外消旋化　　C. 构型翻转　　D. 部分消旋

13. 3，4-二溴代吡啶与等摩尔甲醇钠反应生成

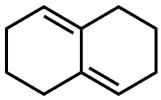
A. 

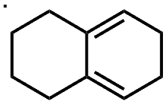
B. 

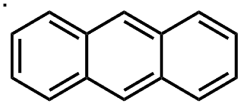
C. 

D. 

14. 下列化合物中，不能发生[4+2]环加成的是

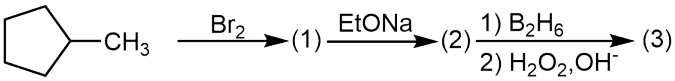
A. 

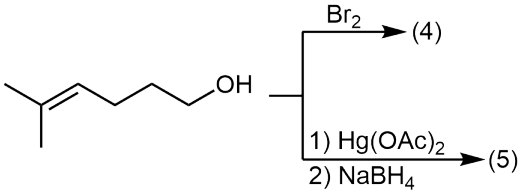
B. 

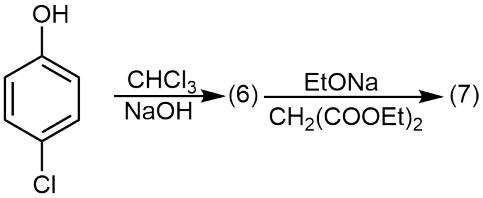
C. 

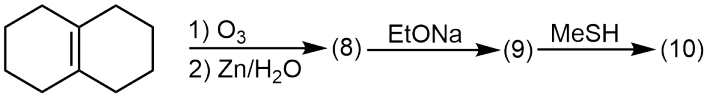
D.

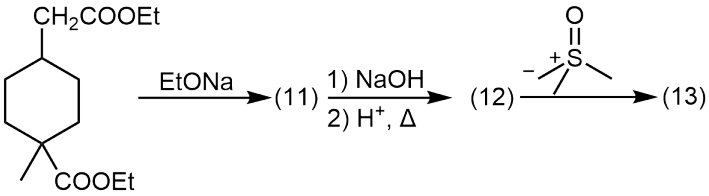
二、完成反应(30 pts.)

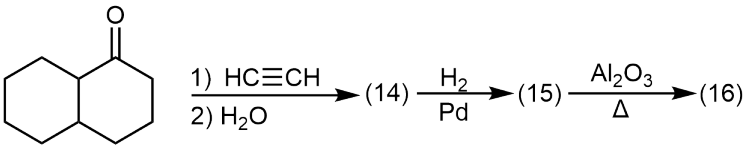
1. 

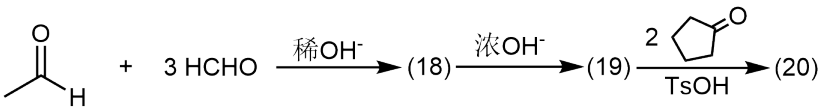
2. 

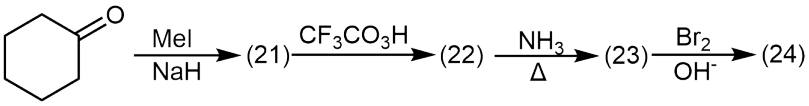
3. 

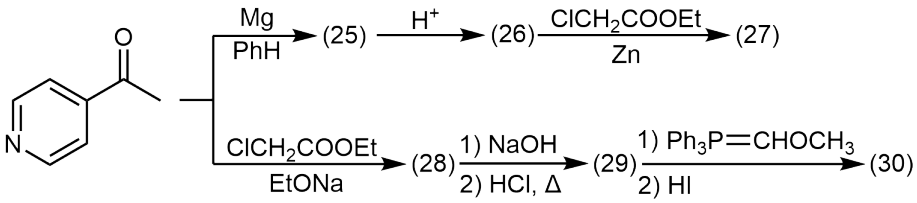
4. 

5. 

6. 

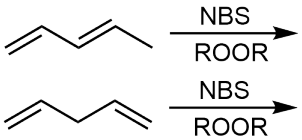
7. 

8. 

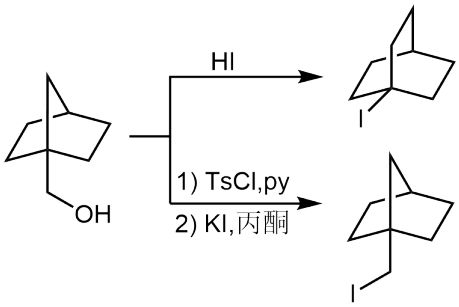
9. 

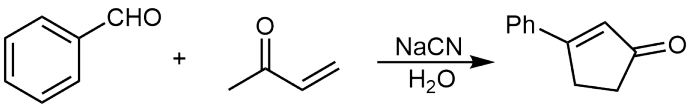
三、简答题(7 pts.)

1. 完成下列反应，并比较哪个反应速度快，说明原因。

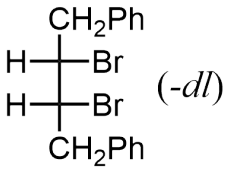


四、机理题(14 pts.)

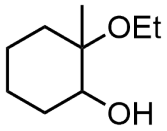
1. 

2. 

五、合成题(35 pts.)



A



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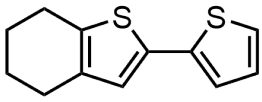

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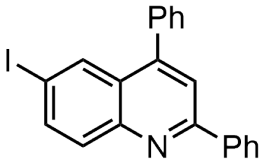
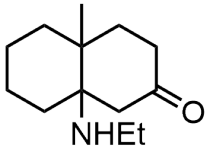


D



E

1. 用不超过两个碳的化合物和苯合成**A**
2. 用不超过三个碳的化合物合成**B**
3. 用环己醇、噻吩、不超过两个碳的化合物合成**C**
4. 用不超过两个碳的化合物和苯合成**D**
5. 用环己醇和不超过三个碳的化合物合成**E**

分析化学部分(共50分)

六、简答题

1. 以某吸附指示剂($\mathrm{p}K_\mathrm{a} = 7.0$)做银量法的指示剂，测定的pH应控制在什么范围？
2. 用重铬酸钾法测定铁矿石中铁的含量，进行预处理时通常加入什么还原物质？为什么要加入H_3PO_4？
3. 写出下列的质子条件式：
   (a) 将20 mL 0.10 mol/L NaOH和10 mL 0.10 mol/L H_2SO_4的溶液混合；
   (b) 将20 mL 0.10 mol/L HCl溶液和20 mL 0.10 mol/L Na_2CO_3的溶液混合。
4. 对某试样进行四次分析结果如下：29.03、29.08、28.97、29.24（%），计算平均值的置信区间。已知$t_{0.90,3} = 2.35$。

5. 向20 mL 0.10 mol/L Ce^{4+}溶液中分别加入20 mL、25 mL 0.1000 mol/L $FeCl_2$溶液，平衡时体系的电位分别为多少？（其中$\varphi^{\ominus}_{Ce^{4+}/Ce^{3+}} = 1.44\,V$, $\varphi^{\ominus}_{Fe^{3+}/Fe^{2+}} = 0.68\,V$）

七、计算题

1. 某弱酸HA总浓度为2.0×10^{-4} mol/L。于520 nm处，用1 cm比色皿测定，在不同pH值的缓冲溶液中，测得吸光度值A如下：

pH	0.88	1.17	2.99	3.41	3.95	4.89	5.50
A	0.890	0.890	0.692	0.552	0.385	0.260	0.260

根据公式 $pK_a = pH + \lg\frac{A-A(A^-)}{A(HA)-A}$ ，其中$A(HA)$和$A(A^-)$为酸和碱型体的吸光度，求：

(a) HA的pK_a；

(b) 用0.02 mol/L的NaOH滴定20.00 mL的0.02 mol/L HA与0.04 mol/L H_3BO_3的混合溶液，滴定终点的pH为6.5，求化学计量点的pH及终点误差。（H_3BO_3的$pK_a = 9.24$）

2. 试计算证明用沉淀掩蔽法在pH=12时用EDTA能准确滴定Ca^{2+}, Mg^{2+}混合溶液中的Ca^{2+}，而Mg^{2+}不干扰。

[已知Ca^{2+}, Mg^{2+}及EDTA浓度均为0.020 mol/L，$Mg(OH)_2$的$pK_{sp} = 10.7$，$\lg K_{CaY} = 10.7$，$\lg K_{MgY} = 8.7$，pH=12时$\lg \alpha_{Y(H)} = 0$]

3. 求AgCl沉淀在pH=8的HL溶液中的溶解度。其中HL的总浓度为0.10 mol/L，$K_{sp}(AgCl) = 1.8\times10^{-10}$，$Ag^+$与$L^-$络合物的$\lg\beta_1 = 3.0$, $\lg\beta_2 = 7.0$，且$K_a(HL) = 1.0\times10^{-10}$。

2025有机化学与分析化学

时量180分钟　满分150分

注意事项:

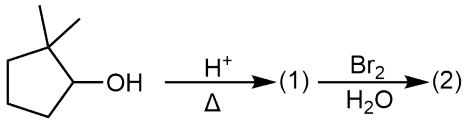
1. 答题前,考生先将自己的姓名、学号、考场号、座位号填写在答题卡上,将条形码准确粘贴在考生信息条形码粘贴区。
2. 考试结束后,将本试卷与答题纸一并收回。
3. 回答各个问题时,将答案写在答题纸上,写在本试卷上无效。

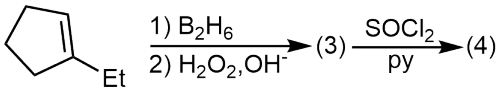
有机化学部分(共100分)

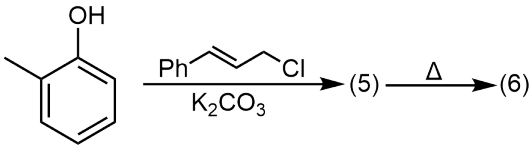
一、选择题（15 pts.）

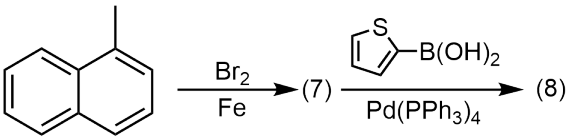
1. 缺失

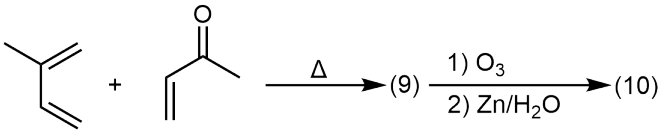
二、完成反应(29 pts.)

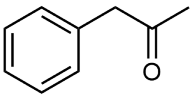
1. 

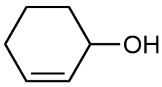
2. 

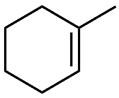
3. 

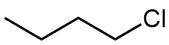
4. 

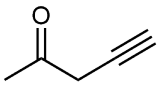
5. 

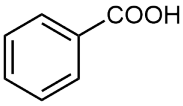
6. 

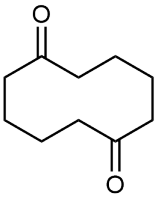
7. 

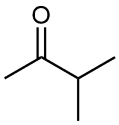
8. 

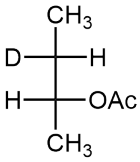
9. 

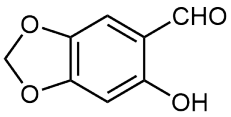
10. 

11. 

12. 

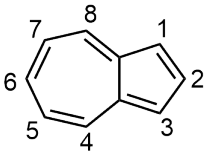
13. 

14. 

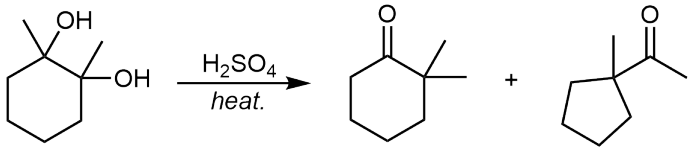
15. 

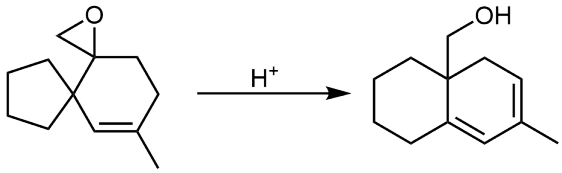
三、简答题(7 pts.)

1. 解释薁(Azulene)在 1-位发生亲电取代反应，在 4,6-位发生亲核取代反应的原因.

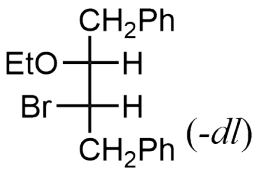


四、机理题(14 pts.)

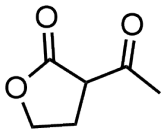
1. 

2. 

五、合成题(35 pts.)



A



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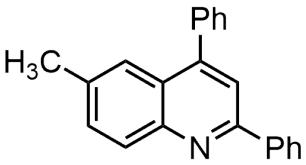

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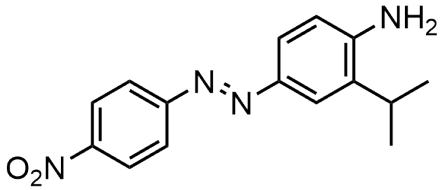
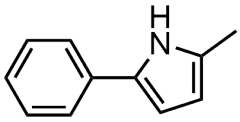


D



E

1. 以苯和乙炔为主要有机原料合成化合物**A**

2. 以不超过两个碳原子的有机原料合成化合物**B**

3. 以苯和不超过两个碳原子的有机原料合成化合物**C**

4. 以苯和不超过三个碳原子的有机原料合成化合物**D**

5. 以苯不超过三个碳原子的有机原料合成化合物**E**

分析化学部分(共50分)

六、简答题：简要回答下列问题。本题共 5 小题，每小题 4 分，共 20 分。

1. 标定盐酸时使用以下两种基准物质，无水碳酸钠，十水合硼砂，据此回答下列问题并解释原因：
   (1) 请判断以下情况对盐酸的标定有何影响？
      a. Na_2CO_3 中吸收少量水分
      b. 硼砂失去少量结晶水
   (2) 若不存在上述误差，且操作过程完全正确，标定盐酸时，用何种物质较好？

2. 根据 $K_2Cr_2O_7$ 法测定铁含量的实验回答下列问题：
   (1) 加入稍过量 $SnCl_2$ 的目的是什么？为什么不能加入太多 $SnCl_2$？
   (2) 体系中使用硫磷混酸的作用是什么？
   (3) 实验使用了指示剂二苯胺磺酸钠，该指示剂氧化态和还原态的颜色分别为？

3. EDTA 可以连续测定样品中的 Bi^{3+} 和 Pb^{2+} 含量。选用二甲酚橙(XO)为指示剂，先在 $pH = 1$ 介质中滴定 Bi^{3+}，然后加入六亚甲基四胺调节 pH 至 5，滴定 Pb^{2+}。已知 $\lg K_{BiY}$ 和 $\lg K_{PbY}$ 分别为 27.94 和 18.04。
   (1) 滴定 Bi^{3+} 时酸度过高或过低对结果有何影响？
   (2) 为什么要调节 pH 至 5？能否使用 NaAc 或者 $NH_3 \cdot H_2O$ 的缓冲溶液？
   (3) 全滴定过程中体系的颜色如何变化？分别对应哪一阶段？

4. （判断有无系统误差的题，用 $t = \frac{|\mu - \bar{x}|}{s}\sqrt{n}$，均值和方差都已给出，不用计算）

5. 根据下表所给数据，填写下列表格中的空缺值：

滴定体系	计量点前 1%	计量点	计量点后 1%
Ce^{4+} 滴定 Fe^{2+}	0.86	1.06	
Fe^{3+} 滴定 Sn^{2+}		0.32	0.50

七、解答题：解答应给出详细步骤。本题共 3 小题，每小题 10 分，共 30 分。

1. 用 0.20 mol/L 的 NaOH 溶液滴定同浓度的甲酸(HA)溶液，已知 $pK_a(HA) = 3.74$。
   (1) 求化学计量点的 pH 及滴定突跃范围。
   (2) 当 $pH_{ep} = 7.0$ 时，计算终点误差。

2. 计算 $PbSO_4$ 在纯水以及 1 mol/L 的 HCl 溶液中的溶解度。
   已知：H_2SO_4 的 $pK_{a2} = 2$；Pb−Cl 配合物的 $\lg\beta_1 \sim \lg\beta_3$ 分别为 1.2、0.6、1.2；$PbSO_4$ 的 $\lg K_{sp} = -8.0$；不考虑盐效应。

3. 在 pH = 5.5 的缓冲溶液中，用 0.020 mol/L 的 EDTA 标准溶液滴定含有 0.020 mol/L 的 Al^{3+} 和 Pb^{2+} 混合液中的 Pb^{2+}，采用二甲酚橙(XO)作为指示剂。在溶液中加入固体 NH_4F（假定 pH 保持不变），至平衡时 $[F^-] = 0.01$ mol/L，计算确定掩蔽效果以及终点误差。
   已知：$\lg K_{AlY} = 16.3$，$\lg K_{PbY} = 18.0$；pH = 5.5 时，$pPb_{ep}(XO) = 7.6$，$\lg\alpha_{Y(H)} = 5.5$，$\alpha_{Al(OH)} = 0.4$；Al−F 配合物的 $\lg\beta_1 \sim \lg\beta_6$ 分别为 6.1、11.2、15.0、17.7、19.4、19.7。

2026有机化学与分析化学

时量180分钟　满分150分

注意事项:

1. 答题前,考生先将自己的姓名、学号、考场号、座位号填写在答题卡上,将条形码准确粘贴在考生信息条形码粘贴区。
2. 考试结束后,将本试卷与答题纸一并收回。
3. 回答各个问题时,将答案写在答题纸上,写在本试卷上无效。

有机化学部分(共100分)

一、选择题(15 pts.)

1. (R)-丁-2-醇反应得到(S)-2-溴丁烷的反应机理是

A. S_N1　B. E1　C. E2　D. S_N2

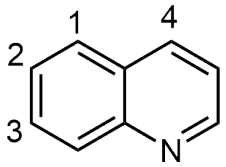
2. 下列化合物中，酸性最强的是

A. $MeOCH_2COOH$　B. CH_3CH_2COOH　C. Cl_3CCOOH　D. $ClCH_2COOH$

3. 下列化合物中，硝化反应速率最快的是

A. 甲苯　B. 硝基苯　C. 吡啶　D. 噻吩

4. 喹啉的亲电取代反应优先进入的位点是

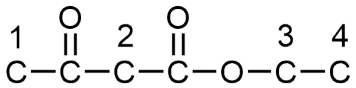


A. 1　B. 2　C. 3　D. 4

5. 苄基甲基醚对下列哪个试剂不稳定

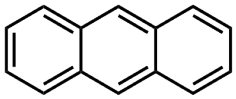
A. 丁基锂　B. Grignard试剂　C. 碱　D. 酸

6. 如下化合物中，酸性最强的氢位于哪个碳上

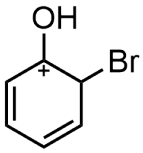
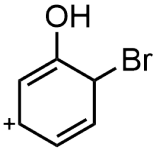
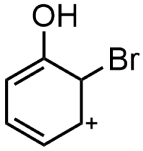
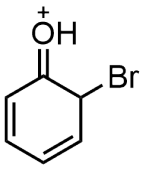


A. 1　　B. 2　　C. 3　　D. 4

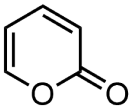
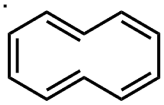
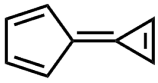
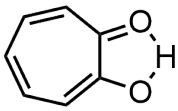
7. 下列共轭烯烃中，不能作为双烯体与乙烯发生[4+2]环加成的是

A. 
B.
C.
D.

8. 下列极限共振式中，对共轭体系贡献最大的是

A. 
B. 
C. 
D. 

9. 下列化合物不具有芳香性的是

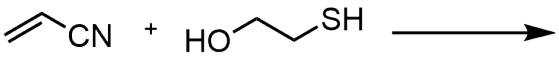
A. 
B. 
C. 
D. 

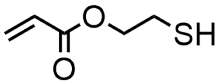
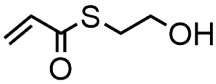
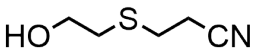
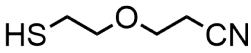
10. 下列化合物中，烯醇式含量最高的是

A. CH_3COCH_2CHO　　B. $CH_3COCH_2COCH_3$

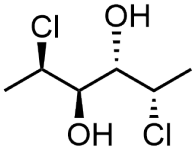
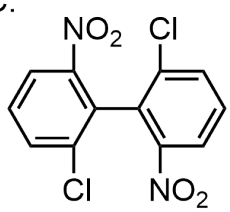
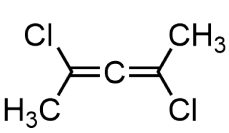
C. $CH_2(COOEt)_2$　　D. CH_3COCH_2COOEt

11. 如下反应的产物是

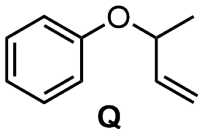


A. 
B. 
C. 
D. 

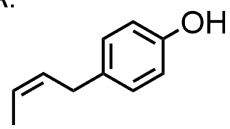
12. 下列化合物中，不具有手性的分子是

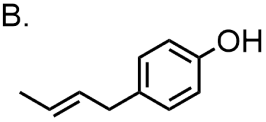
A.
B. 
C. 
D. 

13. 将化合物Q加热到200℃，主产物是



Q

A.


B.


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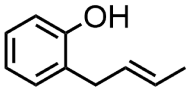
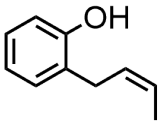
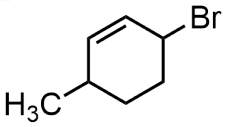
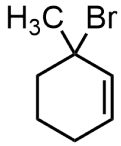
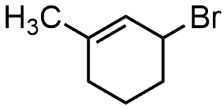


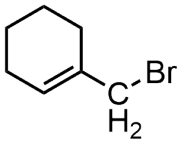
D.


14. 3-甲基环己烯于NBS/ROOR反应条件下在CCl4中回流，主产物是：

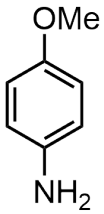
A.


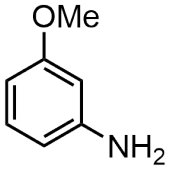
B.


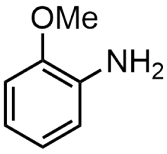
C.


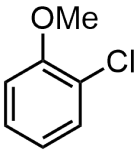
D.


15. 用$NaNH_2/NH_3(l)$处理1-甲氧基-2-氯苯得到

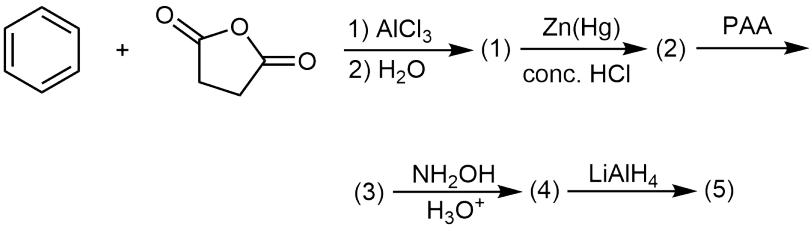
A.


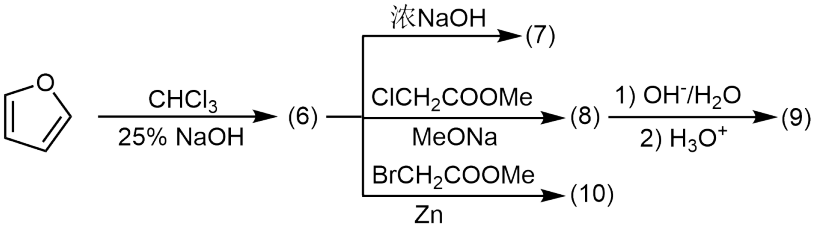
B.


C.


D.


二、完成反应(25 pts.)

1.


2.


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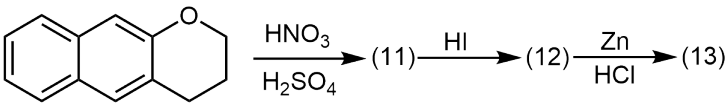
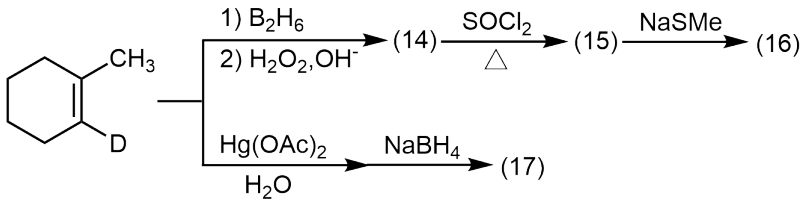
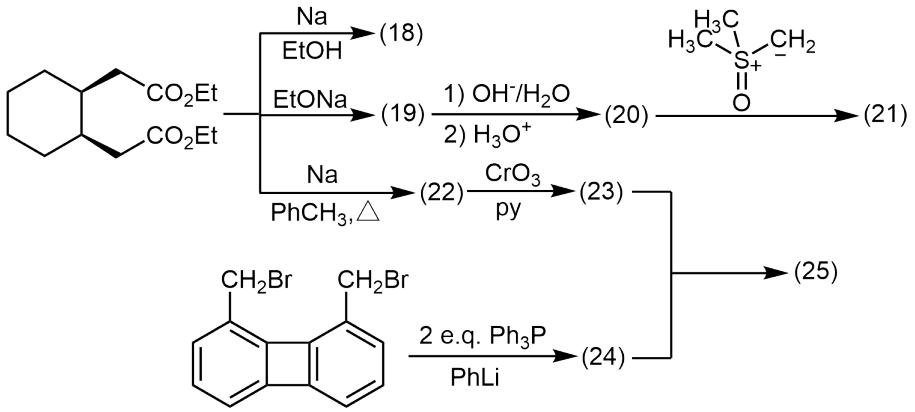
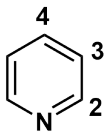


4. 

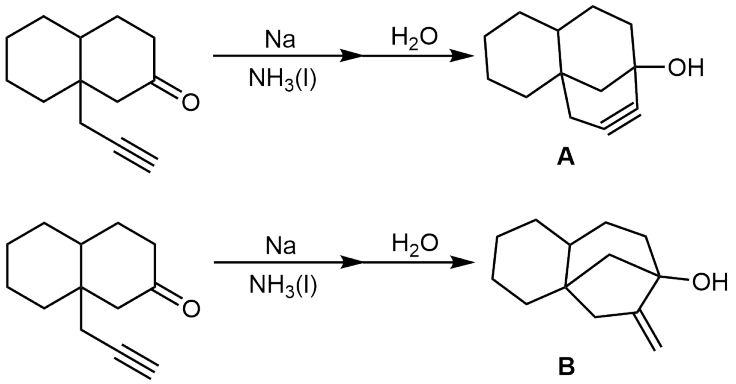
5. 

三、简答题(12 pts.)

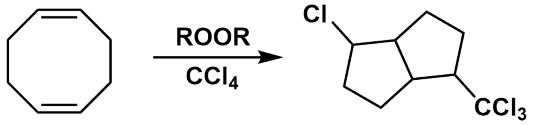
1. 解释吡啶亲电取代取代优先发生在3位、亲核取代优先发生在2、4位的原因

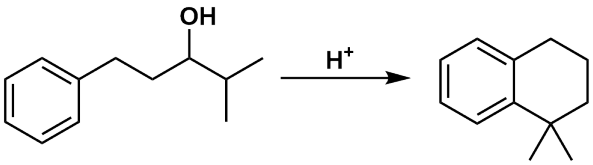


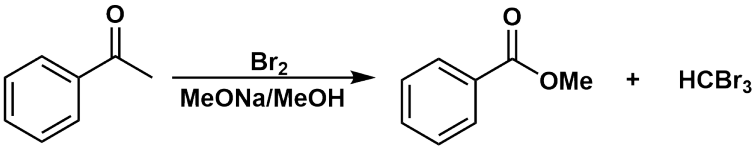
2. 设计如下过程合成A，但未成功，实际发生下方反应得到了B。从目标分子的结构和反应机理两个角度论述原因。



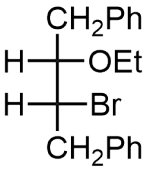
四、机理题(18 pts.)

1. 

2. 

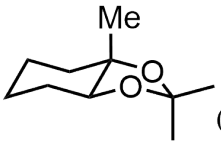
3. 

五、合成题(30 pts.)



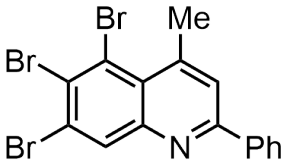
(-*dl*)

A

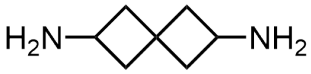


(±)

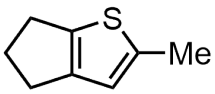
B



C



D



E

1. 以苯、小于等于2个碳的有机物合成**A**
2. 以小于等于3个碳的有机物合成**B**
3. 以苯或甲苯、小于等于3个碳的有机物合成**C**
4. 以小于等于3个碳的有机物合成**D**
5. 以环戊醇、小于等于2个碳的有机物合成**E**

分析化学部分(共50分)

六、选择题(20 pts.)

1. 以某吸附指示剂(p$K_a = 5.0$)作为银量法的指示剂，测定的pH应控制在

A. $pH < 5.0$　　B. $pH > 5.0$　　C. $5.0 < pH < 10.0$ D. $pH > 10.0$

2. 有两组分析数据，若要比较他们的精密度有无显著性差异，则应当用

A. F检验　　B. t检验　　C. u检验　　D. Q检验

3. 含有0.10 mol/L HCl和0.20 mol/L H_2SO_4的混合溶液的质子条件式是

A. $[H^+] = [OH^-] + 2[Cl^-] + [SO_4{}^{2-}]$
B. $[H^+] + 0.3 = [OH^-] + [SO_4{}^{2-}]$
C. $[H^+] - 0.3 = [OH^-] + [SO_4{}^{2-}]$
D. $[H^+] - 0.3 = [OH^-] + [SO_4{}^{2-}] + [HSO_4{}^-]$

4. EDTA与金属离子络合时，一分子的EDTA可提供的配位原子个数是

A. 2　　B. 4　　C. 6　　D. 8

5. 若络合滴定反应为：$M + Y = MY$，则酸效应系数$\alpha_{Y(H)}$的表达式为

A. $[Y]/c(Y)$　　B. $\sum[H_iY]/c(Y)$
C. $[Y]/(\sum[H_iY] + [Y])$　　D. $(\sum[H_iY] + [Y])/[Y]$

6. 用重量法测定试样中的砷，首先使其形成Ag_3AsO_4沉淀，然后转化为AgCl，并以此为称量形，则用As_2O_3表示的换算因数是

A. $M(As_2O_3)/M(AgCl)$　　B. $2M(As_2O_3)/3M(AgCl)$
C. $3M(AgCl)/M(As_2O_3)$　　D. $M(As_2O_3)/6M(AgCl)$

7. 以下银量法测定需采用返滴定方式的是

A. 莫尔法测Cl^-
B. 吸附指示剂法测Cl^-
C. 佛尔哈德法测Cl^-
D. $AgNO_3$滴定CN^-(生成$Ag[Ag(CN)_2]$指示终点)

8. 为了消除0.001000 kg中的非有效数字，将该数值正确地表示为以克(g)为单位的数值结果为

A. 1 g　　B. 1.0 g　　C. 1.00 g　　D. 1.000 g

9. Fe^{3+}与Sn^{2+}反应的平衡常数对数值为
已知：$\varphi^{\ominus}(Fe^{3+}/Fe^{2+}) = 0.77$ V，$\varphi^{\ominus}(Sn^{4+}/Sn^{2+}) = 0.15$ V

A. $(0.77-0.15)/0.059$　　B. $2\times(0.77-0.15)/0.059$

C. $3\times(0.77-0.15)/0.059$　　D. $2\times(0.15-0.77)/0.059$

10. 下列表述中错误的是

A. 由于无定形沉淀颗粒小，为防止沉淀穿滤，应选用致密滤纸（慢速）

B. 微溶化合物的临界值(Q/S)愈大，则愈不容易均相成核

C. 相对过饱和度愈大，分散度愈高

D. 均相成核作用是指构晶离子自发形成晶核

七、填空题(20 pts.)

1. 根据：$\varphi^{\ominus}(Fe^{2+}/Fe)=-0.440$ V、$\varphi^{\ominus}(Sn^{4+}/Sn^{2+})=0.154$ V、$\varphi^{\ominus}(Sn^{2+}/Sn)=-0.136$ V、$\varphi^{\ominus}(Cu^{2+}/Cu^{+})=0.159$ V、$\varphi^{\ominus}(Cu^{+}/Cu)=0.522$ V，判断在酸性溶液中金属铁还原Sn^{4+}时生成_____，而还原Cu^{2+}时则生成_____。

2. 若滴定剂与被测物溶液浓度均增加10倍，NaOH滴定HCl的滴定突跃_____；NaOH滴定HAc的滴定突跃_____。（指pH增加或减少多少单位）

3. 用强酸性阳离子交换树脂分离Na^{+}、K^{+}、Cs^{+}、Rb^{+}、Li^{+}，以0.1 mol/L的HCl溶液洗脱，各个离子流出色谱柱的先后顺序为_____。

4. Fe^{3+}/Fe^{2+}电对电位在加入HCl后会_____；加入邻二氮菲后会_____。（填“增加”、“降低”或“不变”）

5. 实验证明，在较低浓度Na_2SO_4存在下，相比纯水$PbSO_4$的溶解度降低；但当Na_2SO_4的浓度$c\geq 0.2\ \mathrm{mol\cdot L^{-1}}$时，$PbSO_4$的溶解度却增大，这是因为_____。

6. 在弱碱性溶液中用EDTA滴定Zn^{2+}，常使用$NH_3-NH_4{}^{+}$溶液，其作用是_____、_____。

7. H_3PO_4的 $pK_{a1}=2.12$，$pK_{a2}=7.20$，$pK_{a3}=12.36$，则$PO_4{}^{3-}$的 $pK_{b1}=$_____，$pK_{b2}=$_____。

8. EDTA滴定金属M的离子的过程中，介质pH越低，则$\alpha_{Y(H)}$值越_____，K'_{MY}值越_____。（填“大”或“小”）

9. 某人误将参比溶液的透射比调至98%，而不是100%，在此条件下测得有色溶液的透射比为36%，则该有色溶液的正确透射比应为_____。

10. 已知EDTA(H_6Y^{2+})的 $pK_{a1} \sim pK_{a6}$分别为0.9、1.6、2.0、2.67、6.16、10.26，在pH $<$ 1.0的溶液中，EDTA的主要存在形式是_____；pH $= 5.0$时是_____。

八、解答题(10 pts.) 用某分析方法测定某试样中Fe_2O_3的质量分数，重复测定6次，结果分别为：41.0%、42.0%、43.0%、44.0%、43.0%、43.0%。

(1) 在90%置信水平下，采用Q检验法判断是否存在可疑值；

(2) 根据(1)的判断结果，计算Fe_2O_3质量分数的平均值(保留三位有效数字)；

(3) 采用标准方法测得该试样Fe_2O_3质量分数为40.0%，该标准方法的标准偏差为$s = 1.03\%$。在95%置信水平下，判断本方法结果是否存在系统误差。

已知：$Q_{0.10,5} = 0.64$，$Q_{0.10,6} = 0.56$，$Q_{0.10,7} = 0.51$
$t_{0.05,5} = 2.57$，$t_{0.05,6} = 2.45$，$t_{0.10,5} = 2.02$，$t_{0.10,6} = 1.94$